

Project: SolarPulse Analytics (Renewable Energy)

1. Project Planning & Management

- **Project Overview:** A four-week initiative focused on leveraging Big Data and AI to transition solar industry maintenance from reactive to predictive.
- **Project Objectives:** To analyze global and Egyptian solar market trends, develop a high-fidelity Power BI solution, and build a Machine Learning engine for energy yield prediction.
- **Team Size:** 4 members
- **Team-members Name:**

Hassan Abdelzaher: Team Leader and ML Engineer (Random Forest Regressor & Prediction Accuracy).

Mohamed Saaed: Data Analyst (EDA, Correlation Heatmaps, & Feature Selection).

Kirollos Wagdy: Data Architect (SQL Integration & Data Integrity).

Aml Abdelaziz: BI Developer & Coordinator (Dashboard Design & Final Documentation).

Tools Required:

- SQL Server
- Microsoft Excel
- Power BI (Jan 2026 Version).

Dataset: Solar Power Generation Data

1.1 Solar Energy Industry Analysis (Business Context)

Conduct comprehensive research on the renewable energy sector, specifically focusing on solar photovoltaic (PV) systems, covering the following areas:

1.1.1 Global Solar Market Overview

- **Market Size & Growth:** Analysis of the global transition to renewable energy and the compound annual growth rate (CAGR) of solar installments (2020–2025).
- **2030 Net-Zero Targets:** Understanding international climate goals and how solar power is positioned as a primary solution for carbon neutrality.

- **Regional Dominance:** Identifying leading markets (China, EU, USA) and the rapid expansion in the Sun Belt regions (Middle East & Africa).
- **Technological Drivers:** Impact of falling Photovoltaic (PV) hardware costs and the rise of Smart Grid integration.

1.1.2 Egyptian Solar Energy Landscape

- **National Strategic Projects:** Analysis of the **Benban Solar Park** (one of the world's largest) and its impact on the national grid.
- **Vision 2030:** Egypt's government mandate to increase renewable energy contribution to 42% of the total energy mix.
- **Geographical Advantage:** Leveraging Egypt's high solar irradiance levels and the challenges of high-temperature operation.
- **Regulatory Framework:** Understanding Power Purchase Agreements (PPA) and Feed-in Tariffs (FiT) in the Egyptian context.

1.1.3 Competitive & Technical Landscape

- **System Architectures:** Comparing Grid-Tied vs. Off-Grid systems and their data signatures.
- **Inverter Ecosystem:** Identifying top-tier global manufacturers (e.g., Huawei, SMA, Sungrow) and benchmarking their efficiency ratings.
- **Hardware Components:** Analyzing the relationship between PV Panel degradation and Inverter conversion losses.

1.1.4 Industry Dynamics

- **Porter's Five Forces for Solar Utilities:** Analyzing the competitive rivalry in solar farm operation and maintenance (O&M).
- **The Solar Value Chain:** From silicon wafer manufacturing to downstream asset management and real-time monitoring.
- **Digital Transformation:** How Big Data and AI are transitioning the industry from reactive to **Predictive Maintenance**.

1.1.5 Risks, Challenges & Mitigation

- **Soiling & Dust Impact:** The significant production loss caused by dust accumulation (critical for Middle Eastern plants).

- **Thermal Loss:** The inverse relationship between high ambient temperatures and solar cell efficiency.
- **Equipment Downtime:** Economic impact of Inverter failures and "String" disconnects.
- **Data Integrity:** Managing sensor calibration errors and missing timestamps in IoT telemetry.

1.1.6 Key Performance Indicators (KPIs)

Defining the standard industry metrics to be developed in the **Power BI** solution:

Technical & Performance KPIs:

- **Daily/Cumulative Yield (kWh)**
- **Specific Yield (kWh/kWp)**
- **Performance Ratio (PR)**
- **Conversion Efficiency**

Operational & Maintenance KPIs:

- **System Availability (%)**
- **Mean Time Between Failures (MTBF)**
- **Performance Deviation**

Financial & Environmental KPIs:

- **Carbon Offset (CO2 Displacement)**
- **Equivalent Green Impact**

PART 2: DATA PREPARATION & MODELING

2.1 Data Connection & Engineering (SQL & Power Query)

Requirements:

1. **Connect to Data Sources**
 - Use **SQL Server** to import raw Kaggle datasets (Plant 1 & Plant 2 Generation and Weather data).

- Use **Excel** for static metadata (Inverter specifications, Plant locations, and Business Glossary).
- Document the connection strings and the use of SQL Views for initial data merging.

2. Data Quality Assessment (The "Solar" Audit)

- Identify missing signals in AC_POWER or DC_POWER (potential sensor failure).
- Check for **duplicates** in SOURCE_KEY (Inverter IDs) at the same timestamp.
- **Integrity Check:** Ensure every SOURCE_KEY in the Generation table exists in the Inverter Dimension table.
- **Outlier Detection:** Identify unrealistic AMBIENT_TEMPERATURE (> 60°C) or IRRADIATION values that don't match daylight hours.

3. Data Transformation Tasks

- **SQL/Power Query Cleaning:** Handle nulls in YIELD by using interpolation or zero-filling (document the logic).
- **Standardization:** Convert all DATE_TIME columns to a unified ISO format.
- **Advanced Calculated Columns:**
 - Inverter Age: Years since installation.
 - Daylight Status: Flag (Day/Night) based on IRRADIATION > 0.
 - Efficiency Bin: Categorize Inverters into (High, Medium, Low) based on AC/DC conversion ratio.
 - Temperature Group: (Cool <25°C, Moderate 25-45°C, High >45°C).

4. Data Type Optimization

- Set DC_POWER and AC_POWER to **Decimal Number** with 3-place precision.
 - Optimize SOURCE_KEY by mapping long strings to **Integer IDs** in a Dimension table to reduce model size.
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2.2 Data Modeling with Power Pivot (Star Schema)

Requirements:

1. Establish Relationships

- Create a **Star Schema** with Fact_Solar_Reading at the center.
- Relationships: Dim_Date, Dim_Inverter, Dim_Plant, and Dim_Weather_Station.
- Verify **1:Many** cardinality from Dimensions to the Fact table.
- Hide Foreign Keys (e.g., Plant_ID, Inverter_ID) from the Report View to keep it clean.

2. Calculation Groups (Advanced Time Intelligence)

- Create a Calculation Group for **Solar Metrics**: (Current, YTD, MTD, Previous Day, Daily Growth %).
- This allows switching between "Total Yield" and "Average Irradiance" using the same time logic.

3. Model Optimization

- Mark Dim_Date as the official Date Table.
 - Organize measures into **Display Folders**: [01_Power_Metrics], [02_Efficiency], [03_Weather], [04_Financials].
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2.3 DAX Measures Development

CRITICAL PERFORMANCE REQUIREMENT: For the "**Performance Ratio (PR)**" calculation (which requires iterating over daily Irradiance vs. Yield):

1. Implement using an **Iterator** (SUMX over Dim_Date).
2. Implement using a **Pre-calculated column** in SQL (Aggregated Daily Table).
3. Compare performance using **DAX Studio** and document which is faster for a multi-million row dataset.

Measure Categories:

1. Power Generation Measures:

- Total DC Power (kW) & Total AC Power (kW)

- Daily Yield (kWh): The amount of energy produced in a single day.
- Cumulative Yield: Total lifetime energy produced by the plant/inverter.
- AC/DC Conversion Efficiency %: $(AC\ Power / DC\ Power)$.

2. Weather & Environment Measures:

- Average Irradiance: Solar intensity.
- Max Ambient Temperature.
- Module vs Ambient Delta: Difference between panel temp and air temp (identifies overheating).

3. Performance & Quality (KPIs):

- **Performance Ratio (PR%)**: $(Actual\ Yield / Theoretical\ Yield\ based\ on\ Irradiance)$.
- **Specific Yield (kWh/kWp)**: Energy produced per unit of installed capacity.
- **Inverter Downtime (Minutes)**: Count of timestamps with zero AC power during daylight.
- **Underperforming Inverters**: Identification of units performing >10% below plant average.

4. Financial & Sustainability:

- Estimated Revenue: $(Daily\ Yield * Feed-in\ Tariff\ rate)$.
- CO2 Offset: $(Total\ Yield * Carbon\ Emission\ Factor)$ - showing environmental impact.
- Revenue Loss: Potential money lost due to inverter downtime.

5. Time Intelligence:

- Yield YTD (Year to Date).
- Same Period Last Month (SPLM).
- Yield WoW % (Week over Week Growth).

PART 3: ROLE-SPECIFIC BUSINESS QUESTIONS

3.1 CHIEF EXECUTIVE OFFICER (CEO)

Strategic Focus: Total energy portfolio health, sustainability goals, and long-term investment viability. **Business Questions:**

1. **Portfolio Performance:** What is our total energy yield (MWh) across all plants, and how has it evolved year-over-year?
2. **Sustainability Targets:** Are we on track to meet our Carbon Offset (CO₂ displacement) targets for the current fiscal year?
3. **Revenue & Growth:** What is the total estimated revenue from energy sales, and what is the growth rate compared to the previous period?
4. **Strategic Risk:** Which plants are consistently performing below their theoretical "Performance Ratio" (PR) benchmark?

3.2 CHIEF FINANCIAL OFFICER (CFO)

Strategic Focus: Revenue optimization, maintenance costs, ROI, and financial forecasting. **Business Questions:**

1. **Revenue Analysis:** What is the monthly and quarterly revenue trend? How much revenue is generated per kW of installed capacity?
2. **Loss Analysis:** What is the financial impact of "System Downtime"? How much revenue are we losing due to equipment failures?
3. **Operational Costs:** What is the estimated cost of maintenance vs. the revenue gained from improved efficiency?
4. **Forecasting:** Based on historical weather patterns and plant performance, what is the projected revenue for the next quarter?

3.3 CHIEF OPERATING OFFICER (COO)

Strategic Focus: Grid stability, operational uptime, asset utilization, and resource allocation. **Business Questions:**

1. **Plant Uptime:** What is the "System Availability" percentage for each plant? Are we meeting the 98% industry uptime standard?
2. **Efficiency Monitoring:** Which plants show the highest "Conversion Efficiency" (DC to AC), and where are the bottlenecks?

3. **Capacity Utilization:** What is our Capacity Utilization Factor (CUF) across different seasons?
4. **Grid Integration:** How much energy is being clipped (curtailment) during peak production hours?

3.4 HEAD OF OPERATIONS & MAINTENANCE (O&M)

Strategic Focus: Equipment health, predictive maintenance, and technical performance. **Business Questions:**

1. **Inverter Health:** Which specific Inverters (SOURCE_KEY) have the highest failure rates or frequent "Zero-Output" events?
2. **Soiling & Cleaning:** How does dust accumulation (Soiling) affect the yield? Can we identify the optimal cleaning cycle based on production degradation?
3. **Thermal Performance:** What is the correlation between MODULE_TEMPERATURE and power loss? Which inverters are prone to overheating?
4. **Sensor Accuracy:** Are there discrepancies between IRRADIATION sensors and actual power output that suggest sensor calibration issues?

3.5 SUSTAINABILITY & COMPLIANCE OFFICER

Strategic Focus: Environmental impact, regulatory compliance, and green certificates. **Business Questions:**

1. **Environmental Impact:** What is the total metric tons of CO2 avoided by our solar operations this year?
2. **Equivalent Impact:** How many trees planted or cars removed from the road does our clean energy production represent? (For marketing purposes).
3. **Regulatory Reporting:** Are our plants operating within the specific power factor and voltage limits required by the National Grid?

3.6 ASSET MANAGER / PROJECT MANAGER

Strategic Focus: Individual asset performance, life-cycle management, and equipment benchmarking. **Business Questions:**

1. **Asset Benchmarking:** How does Plant 1 perform compared to Plant 2 under similar weather conditions?
2. **Degradation Tracking:** Are we seeing a year-over-year decline in panel efficiency beyond the manufacturer's 0.5% guarantee?

3. **Supplier Quality:** Which Inverter brands/models in our portfolio are delivering the best ROI and lowest maintenance needs?

PART 4: DASHBOARD DEVELOPMENT

4.1 Executive Portfolio Dashboard (Strategic Level)

- **Primary Audience:** CEO, Board of Directors, Investors.
- **Requirements:** High-level KPIs focusing on total energy output, sustainability impact, and strategic growth.
- **Must Include:**
 - **Total Energy Yield (MWh):** Current vs. Target.
 - **Performance Ratio (PR%):** The global health score of the portfolio.
 - **Environmental Impact:** Metric tons of CO2 avoided (Carbon Offset).
 - **Revenue vs. Forecast:** Financial health of the solar assets.
 - **Geographic Map:** Performance of different solar plants (Plant 1 vs. Plant 2).
 - **Alerts:** Underperforming plants or critical system outages.

4.2 Financial & Yield Analysis Dashboard

- **Primary Audience:** CFO, Financial Analysts.
- **Requirements:** Focus on ROI, revenue leakage due to downtime, and cost-benefit of maintenance.
- **Must Include:**
 - **Revenue Waterfall:** Breakdown of revenue by plant and inverter group.
 - **Downtime Cost Analysis:** Financial loss calculated from "Zero-Output" periods.
 - **Levelized Cost of Energy (LCOE):** Efficiency of the investment over time.
 - **Maintenance ROI:** Impact of panel cleaning/repair on revenue uplift.
 - **Budget vs. Actual:** Operating expenses (O&M) vs. energy sales.

4.3 Operations & Grid Dashboard

- **Primary Audience:** COO, Grid Operators.
- **Requirements:** Focus on real-time uptime, grid stability, and capacity utilization.
- **Must Include:**
 - **System Availability %:** Percentage of time the plants were connected to the grid.
 - **Capacity Utilization Factor (CUF):** Seasonal analysis of plant usage.
 - **Peak Power Analysis:** Distribution of power output during daylight hours.
 - **Grid Curtailment:** Energy lost when the grid cannot accept more power.
 - **Daily Power Curve:** AC Power vs. Time of Day.

4.4 Maintenance & Asset Health Dashboard

- **Primary Audience:** O&M Managers, Technical Engineers.
- **Requirements:** Granular view of equipment health and predictive maintenance triggers.
- **Must Include:**
 - **Inverter Performance Matrix:** Ranking inverters from most to least efficient.
 - **Thermal Correlation:** Scatter plot of Module Temperature vs. DC Power.
 - **Soiling Analysis:** Tracking degradation trends to trigger panel cleaning.
 - **Failure Log:** Frequency and duration of inverter disconnects.
 - **Sensor Calibration Check:** Comparing Irradiation readings across sensors.

4.5 Predictive Analytics Dashboard (The AI Engine Interface)

- **Primary Audience:** Data Scientists, Strategic Planners.
- **Requirements:** Visualizing the Machine Learning model results (from your Python step).
- **Must Include:**
 - **Actual vs. Predicted Yield:** Line chart showing the accuracy of the ML model.

- **Feature Importance:** Which factors (Temperature, Irradiance, Time) affect production most?
- **Next 24h Forecast:** Predicted energy production based on weather forecasts.
- **Residual Analysis:** Where and why did the model miss the target?

Submission Deadlines (Graduate Track):

- **Implementation:** 01-04-2026
- **Final Presentation:** 30-04-2026